

V0 | SYNTHETIC TRAINING ARTIFACT

LumenTrail Financial

LumenTrail Pathway Installment Loan Underwriting Policy Manual

Version: V0

Effective period: 2026-03-01 through 2026-08-31

Policy status: Historical

Document family: Underwriting manual

This document is completely fictional. All lender names, product names, thresholds, codes, decisions, sources, and operating logic are invented for retrieval and evaluation testing. No real lender policy, legal threshold, or production rule is represented.

Controlled corpus notice: numeric codes are lender-scoped. A code used here may intentionally have a different meaning in another synthetic lender manual.

PRODUCT OVERVIEW

The LumenTrail Pathway Installment Loan is a fictional fixed-payment consumer product offered through Direct and Verified Partner channels. The policy separates Eligibility Gate, Final Underwriting, and Funding Readiness. Every decision event stores channel, stage, evidence timestamp, rule code, and policy version. Version 0 is historical for the effective period shown on the cover.

LumenTrail terminology is lender-specific. Decline is a terminal underwriting disposition. Evidence Required pauses processing for identified information. Specialist Review requires authorized human assessment. Operational Notice communicates workflow status but is not a credit decision. A Pass permits movement to the next gate and is never, by itself, approval or funding authority.

Cross-lender warning

LumenTrail numeric codes are scoped to LumenTrail. Codes 105, 114, 120, and 128 intentionally have different meanings in the NovaCred manual. A code-only question that does not identify the lender is ambiguous.

CHANNELS AND APPLICATION INTAKE

Table 1: Channel Eligibility

Channel	Entry requirement	Required lineage
Direct	Applicant starts from a LumenTrail-controlled application surface	Intake timestamp, source surface, consent event, policy version
Verified Partner	Partner is active and its attestation is no older than 30 days	Partner ID, attestation version, campaign ID, transfer timestamp

A Verified Partner attestation is current for 30 days in V0. An expired or missing attestation fires Code 116 and enters Evidence Required. Partner status does not establish borrower eligibility and cannot override identity, affordability, fraud, compliance, or funding controls.

APPLICANT ROLES

Table 2: Applicant Roles

Role	Definition	Underwriting treatment
Primary Borrower	Person who establishes the repayment obligation	Required; all borrower controls apply
Co-borrower	Second contractual borrower with separate identity and income	Identity controls are individual; affordability may be combined
Household Contributor	Non-borrower whose documented contribution may support residual-income analysis	Does not become an obligor or satisfy borrower identity
Authorized Funding Party	Person authorized to provide a funding destination without becoming a borrower	Ownership and authorization must reconcile; does not satisfy credit rules

ELIGIBILITY GATE

Eligibility Gate checks identity, channel, basic credit availability, income-source consistency, exposure, and early fraud signals. Baseline requirements include age 21, normalized monthly income of at least \$2,200, no active LumenTrail exposure above \$5,000, no 60+ day delinquency in the prior 120 days, and no more than two Direct applications in 30 days.

DIRECT UNDERWRITING RULES

Table 3: Direct Eligibility Rules

Rule	Source	Action	Code
Active Identity Block	TrailID Registry	Decline	101
Sanctions Status	Lumen Compliance Hub	Decline	102
Age or Residency Conflict	Civic Match Service	Evidence Required	103
Income Source Mismatch	Pathway Income Graph	Evidence Required	105
Thin Credit File [1]	NorthArc Bureau	Evidence Required	106
Affordability Floor	Lumen Residual Model	Decline	107
Application Velocity	Lumen Application Ledger	Specialist Review	108
Existing Exposure	Lumen Account Ledger	Decline	109
Bureau Freeze	NorthArc Bureau	Evidence Required	110
Recent Delinquency	NorthArc Bureau	Decline	111
Duplicate Application	TrailID Registry	Specialist Review	112
Shared Device Cluster	Lantern Device Network	Specialist Review	114
Scoring Inputs Incomplete [3]	Lumen Model Service	Evidence Required	115

[1] Thin-file handling is version-specific and defined in Table 10. Any continuation exception clears only the thin-file condition. It does not create a credit score or approve the application.

VERIFIED PARTNER UNDERWRITING RULES

Table 4: Verified Partner Eligibility Rules

Rule	Source	Action	Code
Active Identity Block	TrailID Registry	Decline	101
Sanctions Status	Lumen Compliance Hub	Decline	102
Income Source Mismatch	Pathway Income Graph	Evidence Required	105
Thin Credit File [1]	NorthArc Bureau	Evidence Required	106
Existing Exposure	Lumen Account Ledger	Decline	109
Shared Device Cluster	Lantern Device Network	Specialist Review	114
Partner Attestation	Partner Assurance Register	Evidence Required	116
Partner Source Posture	Partner Assurance Register	Specialist Review	117
Channel Data Mismatch	Lumen Intake Gateway	Evidence Required	118
Campaign Concentration	Lumen Channel Monitor	Specialist Review	119

Version 0 requires an attestation no older than 30 days. Channel evidence is evaluated separately from borrower evidence. A partner can be active while an individual application still fails a borrower control.

FINAL UNDERWRITING AND FUNDING READINESS

Final Underwriting refreshes volatile attributes, evaluates residual income and payment burden, runs the lender's fraud graph, and confirms identity-document validity. Funding Readiness then validates destination, ownership, consent freshness, and final evidence. Passing Final Underwriting is not Ready to Fund until Funding Readiness is complete.

Table 5: Final Underwriting Rules

Rule	Source	Action	Code
Funding Account Ownership	Lumen Payment Graph	Evidence Required [2]	120
Employment Stability	Pathway Income Graph	Evidence Required	121
Cash-Flow Sufficiency	Lumen Cash-Flow Engine	Decline	122
Payment Burden	Lumen Residual Model	Decline	123
Identity Refresh	TrailID Registry	Evidence Required	124
Sanctions Final	Lumen Compliance Hub	Decline	125
Fraud Graph Overlay	Lantern Device Network	Specialist Review	126
Scoring Vector	Lumen Model Service	Evidence Required	127
Identity Document Validity	Lumen Evidence Vault	Evidence Required	128
Address Reconciliation	Civic Match Service	Specialist Review	129
Account Stability	Lumen Cash-Flow Engine	Decline	130
Line Calculation Incomplete	Lumen Decision Service	Evidence Required	131

[2] Code 120 is not a NovaCred routing rule. It is LumenTrail Funding Account Ownership. The permitted continuation path and review authority are defined in Table 10 and the exception register.

[3] When one scoring input is unavailable but all required non-score evidence is intact, an authorized specialist may allow processing to continue to non-score controls. The missing score is not invented, a line cannot be assigned from a blank band, and the continuation is not approval.

Table 6: Funding Readiness Rules

Rule	Source	Action	Code
Destination Mismatch	Lumen Payment Graph	Evidence Required	132
Funding Hold	Lumen Notice Service	Operational Notice	133
Identity Re-verification	TrailID Registry	Evidence Required	134
Consent Stale	Lumen Consent Ledger	Evidence Required	135
Funding Window Notice	Lumen Notice Service	Operational Notice	136
Final Specialist Review	Lumen Review Console	Specialist Review	137
Returned Account Fraud	Lumen Payment Graph	Decline	138
Cash-Flow Refresh	Lumen Cash-Flow Engine	Evidence Required	139

Operational Notice Codes 133 and 136 report funding workflow state. They must coexist with, and must not replace, the actual underwriting or evidence result that produced the notice.

LINE ASSIGNMENT MATRIX

LumenTrail uses a bureau band and an internal Cash-Flow Grade. This differs from NovaCred's matrix. The selected adjustment modifies a synthetic base line only after required controls pass. Missing bands enter Code 131; they are not defaulted.

Table 7: Pathway Line Adjustment Matrix

Bureau band	Grade D	Grade C	Grade B	Grade A
<560	-\$1,200	-\$900	-\$650	-\$400
560-599	-\$900	-\$600	-\$250	+\$100
600-639	-\$550	-\$200	+\$150	+\$300
640-679	-\$250	+\$100	+\$400	+\$600
>=680	\$0	+\$250	+\$600	+\$750

INCOME AND AFFORDABILITY

Normalized monthly income must be at least \$2,200. Final affordability uses residual monthly income after verified obligations and a payment-burden band. A high score cannot override insufficient residual income. Household Contributor income is used only when the relationship and contribution are documented; the contributor does not become a borrower.

Table 8: Income Documentation Bands

Band	Criteria	Required evidence
Standard	Stable salary with current automated confirmation	One current pay statement and automated employment confirmation
Variable	Variable hours, recent employment change, or partial automated history	Two pay statements, employment confirmation, and variance explanation
Independent / Complex	Self-employed, contract, business-derived, or materially disputed income	Three recent bank statements, business-income summary, and independent-income attestation

Income evidence is measured over the window defined by the selected band. A source update creates a new evaluation event. Reviewers retain the stated value, verified value, calculation basis, and reconciliation result.

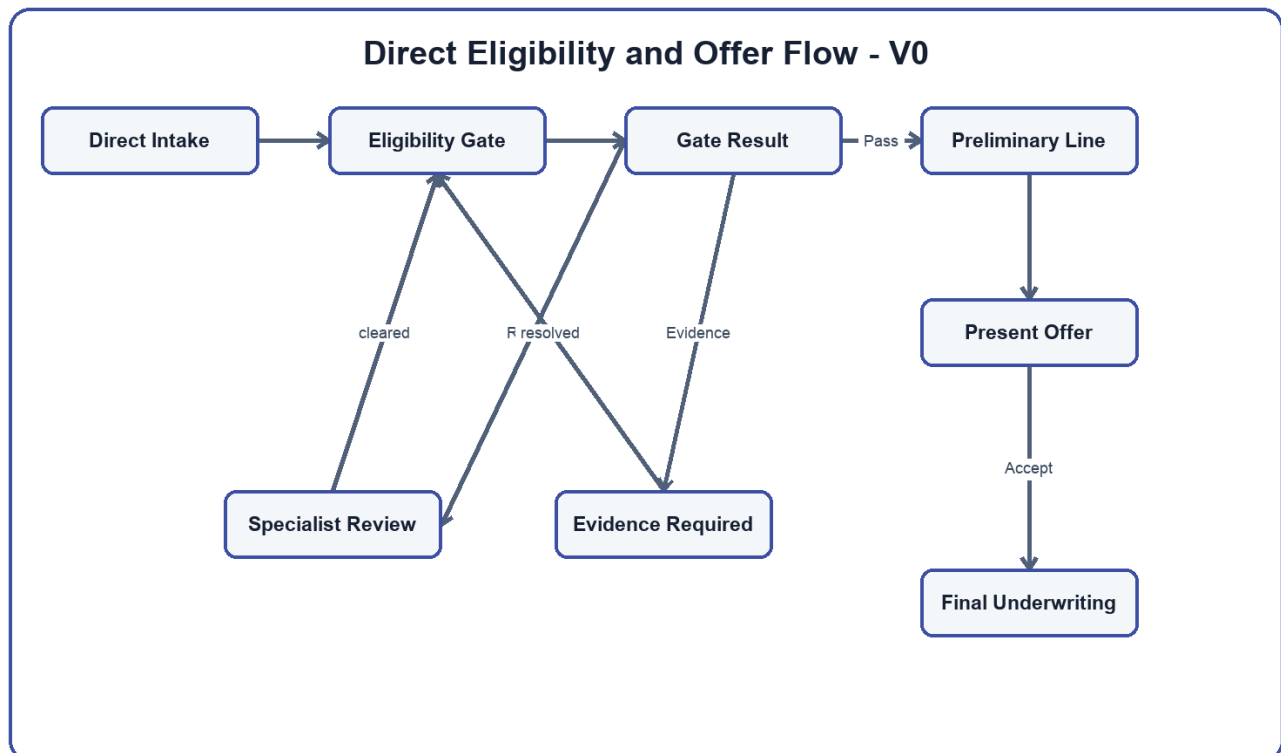
EXHIBIT A - CHANNEL ELIGIBILITY

Direct and Verified Partner applications reach the same Final Underwriting controls, but they enter with different channel evidence. Version 0 partner records require an active partner, a 30-day attestation, and an unbroken transfer lineage. A channel pass only establishes that the application entered through an allowed source.

A missing lender identifier cannot be inferred from the numeric rule code because numeric codes are not globally unique. A missing channel is quarantined. A missing version is resolved from the evaluation date only after lender identity is established.

EXHIBIT B - DECISION FLOWS

Direct Eligibility and Offer Flow - V0



Verified Partner Final Underwriting Flow - V0

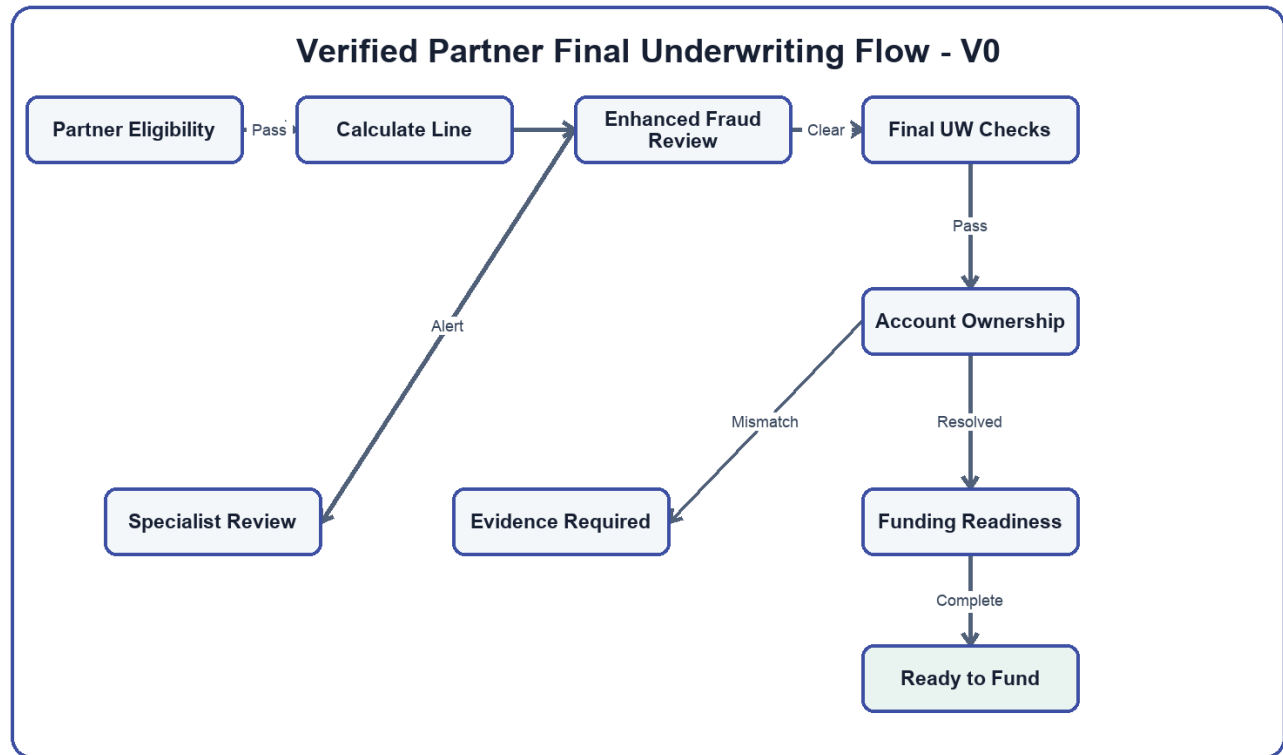


EXHIBIT C - LUMENTRAIL CODE DICTIONARY

This appendix is authoritative for LumenTrail V0. Code meaning is resolved by lender and version before any rule row or threshold is applied.

Table 9: Decline Codes

Code	V0 definition
Decline 101	Active Identity Block: borrower matches an active LumenTrail identity-block record.
Decline 102	Sanctions Status: potential match remains uncleared after permitted comparison.
Decline 107	Affordability Floor: normalized monthly income is below \$2,200 or residual income fails the configured minimum.
Decline 109	Existing Exposure: active LumenTrail exposure exceeds \$5,000.
Decline 111	Recent Delinquency: a 60+ day delinquency appears in the prior 120 days.
Decline 122	Cash-Flow Sufficiency: eligible inflow and residual cash do not meet final requirements.
Decline 123	Payment Burden: projected installment exceeds the permitted payment-burden band.
Decline 125	Sanctions Final: refreshed sanctions status remains uncleared.
Decline 130	Account Stability: eligible cash-flow history fails the final stability threshold.
Decline 138	Returned Account Fraud: confirmed synthetic return pattern meets terminal criteria.

Table 10: Evidence Required Codes

Code	V0 definition
Evidence 103	Age or residency fields conflict across approved sources.
Evidence 105	Income Source Mismatch: stated source and approved evidence do not reconcile.
Evidence 106	Thin Credit File: fewer than two qualifying tradelines requires additional evidence. Version 0 has no automated rent-history continuation exception.
Evidence 110	Bureau Freeze: file access is prevented until a refreshed state is available.
Evidence 115	Scoring Inputs Incomplete: required inputs are missing or cannot be normalized.
Evidence 116	Partner Attestation: evidence is absent or older than 30 days.
Evidence 118	Channel Data Mismatch: transferred partner fields do not reconcile with signed intake.
Evidence 120	Funding Account Ownership: ownership cannot be reconciled automatically, so documentary evidence is required before processing can continue.
Evidence 121	Employment Stability: current status or tenure requires corroboration.
Evidence 124	Identity Refresh: final-stage identity evidence is stale or incomplete.
Evidence 127	Scoring Vector: required normalized vector is missing.
Evidence 128	Identity Document Validity: evidence expires before the expected funding date.
Evidence 131	Line Calculation Incomplete: one or both line-matrix dimensions cannot be established.
Evidence 132	Destination Mismatch: disbursement destination does not reconcile with approved instructions.
Evidence 134	Identity Re-verification: Funding Readiness requires a fresh identity comparison.
Evidence 135	Consent Stale: funding consent is outside its active window.
Evidence 139	Cash-Flow Refresh: final eligible inflow window is stale or incomplete.

Table 11: Operational Notice Codes

Code	Definition
Notice 133	Funding Hold: funding is paused while a separate evidence or review result remains open.
Notice 136	Funding Window Notice: expected disbursement window changed after acceptance.

Table 12: Specialist Review Codes

Code	V0 definition
Review 108	Application Velocity: Direct application count needs behavioral context.
Review 112	Duplicate Application: active records require relationship reconciliation.
Review 114	Shared Device Cluster: device is associated with several applicants; household context may clear it.
Review 117	Partner Source Posture: active source is above its temporary monitoring band.
Review 119	Campaign Concentration: partner campaign exhibits an abnormal application cluster.
Review 126	Fraud Graph Overlay: combined network signals exceed the specialist threshold.
Review 129	Address Reconciliation: approved sources disagree on current residence.
Review 137	Final Specialist Review: an unresolved case requires funding-authority review.

Table 13: Code Cross-Reference Audit

Family	Codes covered	Rule tables
Decline	101, 102, 107, 109, 111, 122, 123, 125, 130, 138	Tables 3-6
Evidence Required	103, 105, 106, 110, 115, 116, 118, 120, 121, 124, 127, 128, 131, 132, 134, 135, 139	Tables 3-6
Operational Notice	133, 136	Table 6
Specialist Review	108, 112, 114, 117, 119, 126, 129, 137	Tables 3-6

Table 14: LumenTrail Risk Signal Catalogue

Signal	Synthetic condition	Default route	Important limitation
LT-R1	Device shared by 5+ unrelated active applications in 18 hours	Review 114	Related household use may be legitimate
LT-R2	Funding account appears across 4+ unrelated applicants	Evidence 120	Ownership evidence controls the final route
LT-R3	Partner transfer fields change after signed intake	Evidence 118	Do not treat as borrower fraud without corroboration
LT-R4	Cash-flow categories change materially after refresh	Evidence 139	Can reflect source recategorization
LT-R5	Identity cluster and returned-account pattern co-occur	Review 126	Terminal action requires the mapped rule
LT-R6	Multiple applicants share a verified household address	Context only	Shared address alone is not adverse

Table 15: Exception Footnote Register

Footnote	Rule	V0 effect
[1]	Thin Credit File	No automated continuation exception in Version 0; request evidence.
[2]	Funding Account Ownership	Documentary evidence may reconcile ownership and permit continuation.
[3]	Scoring Inputs Incomplete	Authorized continuation reaches non-score controls only; no score or line is created.

APPENDIX - EVIDENCE, VERSIONING, AND AUDIT NOTES

SOURCE AND EVIDENCE LINEAGE

Every evaluated value retains raw source, normalized interpretation, observation timestamp, retrieval timestamp, and transformation. Conflicting sources remain addressable. An operator cannot choose a favorable source outside the configured precedence order.

VERSION RESOLUTION

LumenTrail V0 applies to evaluation events from 2026-03-01 through 2026-08-31. The evaluation date, not upload date or PDF modification date, selects the historical policy. Re-evaluation under another version creates a new linked event.

RULE AND SECTION PROVENANCE

Each event records lender, product, document version, rule code, stage, and section identity. Version 0 stores document-level version lineage; it does not require a section revision hash.

REOPENED CASES

A reopened case preserves the earlier disposition and starts a new evaluation event. Version 0 may reuse a prior evidence package when every artifact remains within its configured validity window.

EXCEPTION GOVERNANCE

Exception use records condition, evidence, authorizer, timestamp, and scope. A continuation exception only clears the named rule. It cannot establish approval, calculate a missing line, or bypass Funding Readiness.

MONITORING

Monitoring is segmented by lender, version, channel, stage, source, code, evidence outcome, and final disposition. Cross-lender code counts must never be combined solely by numeric code. Version changes are evaluated against expected threshold and routing effects.

IDENTITY, FRAUD, AND COMPLIANCE CONTROLS

SHARED DEVICE CLUSTERS

Code 114 means LumenTrail Shared Device Cluster. It requires Specialist Review because household or shared-network use can be legitimate. The reviewer compares relationship, identity consistency, timing, and funding ownership. This is not NovaCred FACTA handling despite the shared number.

FUNDING ACCOUNT OWNERSHIP

Funding Account Ownership: ownership cannot be reconciled automatically, so documentary evidence is required before processing can continue. Passing or resolving this rule permits movement to remaining Funding Readiness checks and does not mean the application is approved.

IDENTITY DOCUMENT VALIDITY

Identity Document Validity: evidence expires before the expected funding date. The expected funding date and document expiration date must both remain in the evidence record.

APPLICATION REWORK AND EVIDENCE REQUESTS

EVIDENCE REQUIRED LIFECYCLE

An Evidence Required event identifies the requested artifact, source rule, stage, expiration, and permitted recovery action. When evidence arrives, the active stage reruns under the policy version selected for the new evaluation event. The original request remains in history.

MULTIPLE OPEN REQUESTS

One document may satisfy more than one request, but every source code remains separately traceable. Requests may be consolidated for the applicant without collapsing their audit relationships.

FUNDING HOLDS AND NOTICES

A Funding Hold or Funding Window Notice communicates operational state only. The answer to whether the applicant passed, requires evidence, entered review, or was declined must come from the corresponding underwriting result, not the notice code.